



Mansoura University
Faculty of Computers and Information
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MODELLING & SIMULATION

SW,BI,AI

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COURSE SYLLABUS

- **Chapter 1: Introduction.**
- **Chapter 2: Probability as Using in Simulation.**
- **Chapter 3: Queueing Simulation.**
- **Chapter 4: Inventory Simulation.**
- **Chapter 5: Random-Number Generation.**
- **Chapter 6: Input Modeling.**
- **Chapter 7: Output Analysis.**

COURSE SYLLABUS

- **Chapter 1: Introduction.**
- **Chapter 2: Probability as Using in Simulation.**
- **Chapter 3: Queueing Simulation.**
- **Chapter 4: Inventory Simulation (part I).**
- **Chapter 5: Random-Number Generation.**
- **Chapter 6: Input Modeling.**
- **Chapter 7: Output Analysis.**

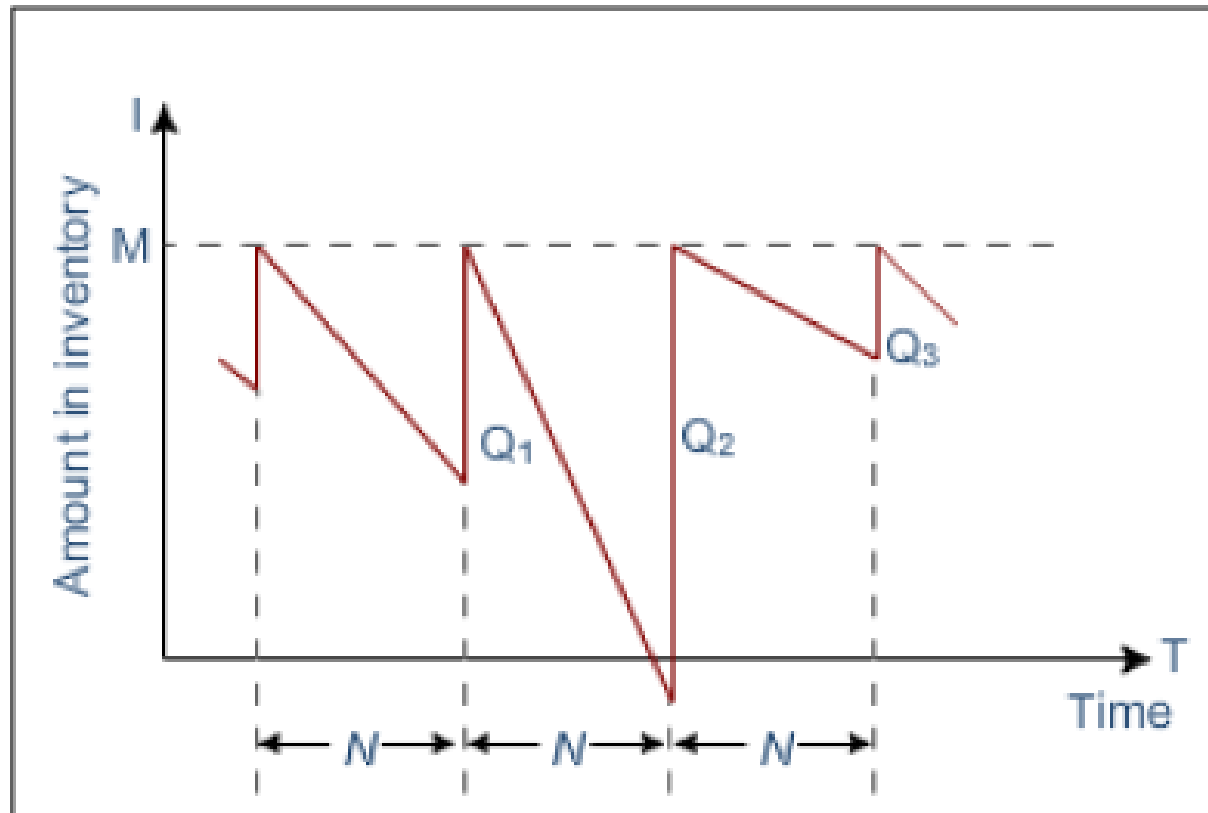
CHAPTER CONTENT: INVENTORY SIMULATION

- Introduction.
- Simulating of Inventory System.
- The Newspaper Seller's Problem.

INTRODUCTION

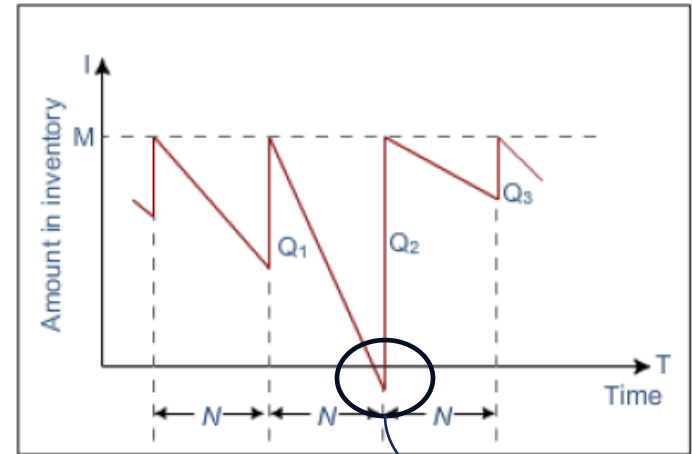
- An **inventory system** represents the stock of items kept on hand to meet future demand.
- Organizations maintain inventory to meet customer demand, buffer against uncertainties, and take advantage of economies of scale in purchasing.

INVENTORY SYSTEM



INVENTORY SYSTEM

- **N** : Periodic review length (months/days).
- **M** : Standard inventory level.
- **Q_i** : Quantity of order i to fill up to M .
 - ✓ In this example, **Lead Time** is zero.
- Since demands are not usually known with certainty, the order quantities are probabilistic.



The length of time between the **placement** and **receipt** of an order.

KEY ELEMENTS OF AN (M, N) INVENTORY SYSTEM

Parameter	Description
N	Periodic review length (months, days, or other time units)
M	Standard inventory level (maximum desired inventory)
Q_i	Quantity of order i to fill inventory up to M

Lead time is the length of time between the placement and receipt of an order. In some systems, lead time may be zero (immediate delivery), while in others it may be several days or weeks.

THE SHORTAGE PROBLEM

- To avoid shortages, a **buffer stock** is needed.
- Carrying stock in inventory has an associated cost:
 - Funds to buy the items,
 - Renting of storage space,
 - Hiring guards.

THE SHORTAGE PROBLEM

- An alternative to carrying high inventory is to make **more frequent reviews**.
- This has an associated cost:
 - The ordering cost.
 - Delivery.
 - Expiry date in some items.

PERFORMANCE MEASURE

- The **total cost** (or total profit) of an inventory system is the measure of performance. The decision maker can control:
 - **M:** The maximum inventory level
 - **N:** The length of the review cycle
- A key question is: **What effect does changing N have on the various costs?**

EVENTS IN AN (M, N) INVENTORY SYSTEM

- The following events occur in sequence:

Event	Description
1. Demand for items	Customers request items, reducing inventory
2. Review of inventory	At the end of each review period, inventory level is checked
3. Order placement	If needed, an order is placed to bring inventory up to M
4. Receipt of order	After lead time, the order arrives and increases inventory

SIMULATION OF AN (M, N) INVENTORY SYSTEM

- The **problem** is to estimate, by simulation, two key performance measures:
 - Average ending units in inventory - This indicates the typical stock level
 - Number of days when a shortage condition occurs - This indicates service level
- These measures help us choose a good combination of **(M, N)**.

To simulate an inventory system, we need:

Demand Distribution	The probability distribution of the number of units demanded per day
Lead Time Distribution	The probability distribution of the lead time (if not constant)

EXAMPLE

- Suppose that the maximum inventory level, M , is 11 units and the periodic review length, N , is 5 days.

M : Standard inventory level = 11 units

N : Periodic review length = 5 days

perform the simulation for 5 cycles.

Initial state:

- ✓ *The inventory level start at 3 units.*
- ✓ *An order of 8 units scheduled to arrive in 2 days' time.*

EXAMPLE

Random-Digit Assignments for Daily Demand

<i>Demand</i>	<i>Probability</i>	<i>Cumulative Probability</i>	<i>Random-Digit Assignment</i>
0	0.10		
1	0.25		
2	0.35		
3	0.21		
4	0.09		

Random-Digit Assignments for Lead Time

<i>Lead Time (Days)</i>	<i>Probability</i>	<i>Cumulative Probability</i>	<i>Random-Digit Assignment</i>
1	0.6		
2	0.3		
3	0.1		

EXAMPLE

Random-Digit Assignments for Daily Demand

<i>Demand</i>	<i>Probability</i>	<i>Cumulative Probability</i>	<i>Random-Digit Assignment</i>
0	0.10	0.10	01–10
1	0.25	0.35	11–35
2	0.35	0.70	36–70
3	0.21	0.91	71–91
4	0.09	1.00	92–00

Random-Digit Assignments for Lead Time

<i>Lead Time (Days)</i>	<i>Probability</i>	<i>Cumulative Probability</i>	<i>Random-Digit Assignment</i>
1	0.6	0.6	1–6
2	0.3	0.9	7–9
3	0.1	1.0	0

EXAMPLE

■ Simulation Table (Cycle I)

<i>Cycle</i>	<i>Day</i>	<i>Beginning Inventory</i>	<i>Random Digits for Demand</i>	<i>Demand</i>	<i>Ending Inventory</i>	<i>Shortage Quantity</i>	<i>Order Quantity</i>	<i>Random Digits for Lead Time</i>	<i>Days until Order Arrives</i>
1	1								
	2								
	3								
	4								
	5								

EXAMPLE

■ Simulation Table (Cycle I)

<i>Cycle</i>	<i>Day</i>	<i>Beginning Inventory</i>	<i>Random Digits for Demand</i>	<i>Demand</i>	<i>Ending Inventory</i>	<i>Shortage Quantity</i>	<i>Order Quantity</i>	<i>Random Digits for Lead Time</i>	<i>Days until Order Arrives</i>
1	1	3	24	1	2	0	–	–	1
	2	2	35	1	1	0	–	–	0
	3	9	65	2	7	0	–	–	–
	4	7	81	3	4	0	–	–	–
	5	4	54	2	2	0	9	5	1

EXAMPLE

■ Simulation Table (Cycle 2)

Cycle	Day	Beginning	Random	Demand	Ending	Shortage	Order	Random	Days until
		Inventory	Digits for		Inventory	Quantity	Quantity	Digits for	Order
								Lead Time	Arrives
2	5	4	54	2	2	0	9	5	1
	1	2	03	0	2	0	—	—	0
	2	11	87	3	8	0	—	—	—
	3	8	27	1	7	0	—	—	—
	4	7	73	3	4	0	—	—	—
	5	4	70	2	2	0	9	0	3

EXAMPLE

■ Simulation Table (Cycle 3)

Cycle	Day	Beginning Inventory	Random Digits for Demand	Demand	Ending Inventory	Shortage Quantity	Order Quantity	Random Digits for Lead Time	Days until Order Arrives
	5	4	70	2	2	0	9	0	3
3	1	2	47	2	0	0	-	-	2
	2	0	45	2	0	2	-	-	1
	3	0	48	2	0	4	-	-	0
	4	9 5	17					-	-
	5		09						

EXAMPLE

■ Simulation Table (Cycle 3)

Cycle	Day	<i>Random</i>		<i>Ending</i>	<i>Shortage</i>	<i>Order</i>	<i>Random</i>		<i>Days until</i>
		<i>Beginning</i>	<i>Digits for</i>				<i>Digits for</i>	<i>Order</i>	
		<i>Inventory</i>	<i>Demand</i>	<i>Demand</i>	<i>Inventory</i>	<i>Quantity</i>	<i>Quantity</i>	<i>Lead Time</i>	<i>Arrives</i>
3	5	4	70	2	2	0	9	0	3
	1	2	47	2	0	0	–	–	2
	2	0	45	2	0	2	–	–	1
	3	0	48	2	0	4	–	–	0
	4	9	17	1	4	0	–	–	–
	5	4	09	0	4	0	7	3	1

EXAMPLE

<i>Cycle</i>	<i>Day</i>	<i>Beginning Inventory</i>	<i>Random Digits for Demand</i>	<i>Demand</i>	<i>Ending Inventory</i>	<i>Shortage Quantity</i>	<i>Order Quantity</i>	<i>Random Digits for Lead Time</i>	<i>Days until Order Arrives</i>
1	1	3	24	1	2	0	—	—	1
	2	2	35	1	1	0	—	—	0
	3	9	65	2	7	0	—	—	—
	4	7	81	3	4	0	—	—	—
	5	4	54	2	2	0	9	5	1
2	1	2	03	0	2	0	—	—	0
	2	11	87	3	8	0	—	—	—
	3	8	27	1	7	0	—	—	—
	4	7	73	3	4	0	—	—	—
	5	4	70	2	2	0	9	0	3
3	1	2	47	2	0	0	—	—	2
	2	0	45	2	0	2	—	—	1
	3	0	48	2	0	4	—	—	0
	4	9	17	1	4	0	—	—	—
	5	4	09	0	4	0	7	3	1
4	1	4	42	2	2	0	—	—	0
	2	9	87	3	6	0	—	—	—
	3	6	26	1	5	0	—	—	—
	4	5	36	2	3	0	—	—	—
	5	3	40	2	1	0	10	4	1
5	1	1	07	0	1	0	—	—	0
	2	11	63	2	9	0	—	—	—
	3	9	19	1	8	0	—	—	—
	4	8	88	3	5	0	—	—	—
	5	5	94	4	1	0	10	8	2
					88				

PERFORMANCE ANALYSIS

Performance Measure	Calculation	Result
Total Ending Inventory	Sum of ending inventory each day	88 units
Average Ending Inventory	Total Ending Inventory ÷ Number of Days	$88 \div 25 = 3.52$ units
Shortage Days	Days with shortage > 0	2 days
Probability of Shortage	Shortage Days ÷ Total Days	$2 \div 25 = 0.08$ (8%)
Total Shortage Quantity	Sum of shortage quantities	6 units
Total Demand	Sum of daily demand	41 units

PERFORMANCE ANALYSIS

- Based on five cycles of simulation, the average ending inventory is approximately 3.5 ($88 \div 25$) units.
- On 2 of 25 days a shortage condition existed.
- For large number of cycles, the computer is used.

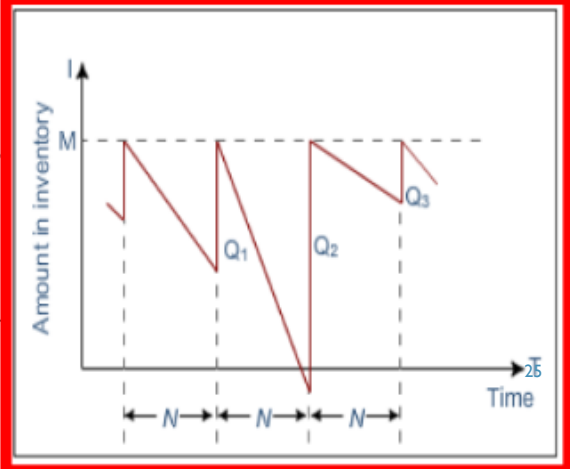
GRAPH

Simulation Table (5 Cycles)

Cycle	Day	Beginning Inventory	Random Digits for Demand	Demand	Ending Inventory	Shortage Quantity	Order Quantity	Random Digits for Lead Time	Days until Order Arrives
1	1	3	24	1	2	0	—	—	1
	2	2	35	1	1	0	—	—	0
	3	9	65	2	7	0	—	—	—
	4	7	81	3	4	0	—	—	—
	5	4	54	2	2	0	—	—	1
2	1	2	03	0	2	0	—	—	0
	2	11	87	3	8	0	—	—	—
	3	8	27	1	7	0	—	—	—
	4	7	73	3	4	0	—	—	—
	5	4	70	2	2	0	9	0	3
3	1	2	47	2	0	0	—	—	—
	2	0	45	2	0	2	—	—	—
	3	0	48	2	0	4	—	—	—
	4	9	17	1	4	0	—	—	—
	5	4	09	0	4	0	—	—	—
4	1	2	—	2	2	0	—	—	—
	2	—	—	3	6	0	—	—	—
	3	6	26	1	5	0	—	—	—
	4	5	36	2	3	0	—	—	—
	5	3	40	2	1	0	—	—	—
5	1	1	07	0	1	0	—	—	—
	2	11	63	2	9	0	—	—	—
	3	9	19	1	8	0	—	—	—
	4	8	88	3	5	0	—	—	—
	5	5	94	4	1	0	—	—	—

Amount: y-Axis

Time: x-Axis





Thanks
Any questions